



Gemini API Developer Competition Submission

Project name: GeminiCare

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Grade: 11th

Overview: [A Gemini powered App assisting medical professionals in diagnosing and monitoring autistic patients.](#)

App link: <http://35.223.243.41:8501/>

Github link: <https://github.com/srijonmandal1/gemini-care-autism-treatment>

User Guide link: <https://tinyurl.com/n6rhp796>

Video link:



Gemini powered App for assisting Medical Professionals to diagnose, monitor and treat Autism patients efficiently.

Problem

- Clinical and scientific data show a higher risk of adverse events and trauma in people with an autism spectrum disorder (ASD).
- There is significant challenges in timely treatment of ASD in non-invasive manner



Treatment Gap



Diagnostic Challenges



Chronic Mental Health



Suicide Risk



Harm to others

AUTISM QUICK STATS



1 in 36 kids identified with autism



241%
higher than baseline
stats in 2000

Although autism can be diagnosed before a child reaches the age of two, most kids are diagnosed only after the age of

4



BOYS are 4X more likely to be diagnosed than **GIRLS**

40% are nonverbal
44% Have average or above average intellectual ability
31% have an intellectual disability

Some areas have higher than average prevalence rates:
California 1/26
South Korea 1/38

36.5% of autism caregivers use ABA

20% of caregivers give CBD to autistic kids

From 4-5 in every 10,000 in the 70s to half of all kids in 2025

Will half of US kids have autism by 2025? This prediction made by Dr. Stephanie Seneff, Research Scientist from the Massachusetts Institute of Technology (MIT), may be scoffed at by some, [but the latest CDC statistics](https://www.autismparentingmagazine.com/autism-statistics/) do indicate rising prevalence rates:

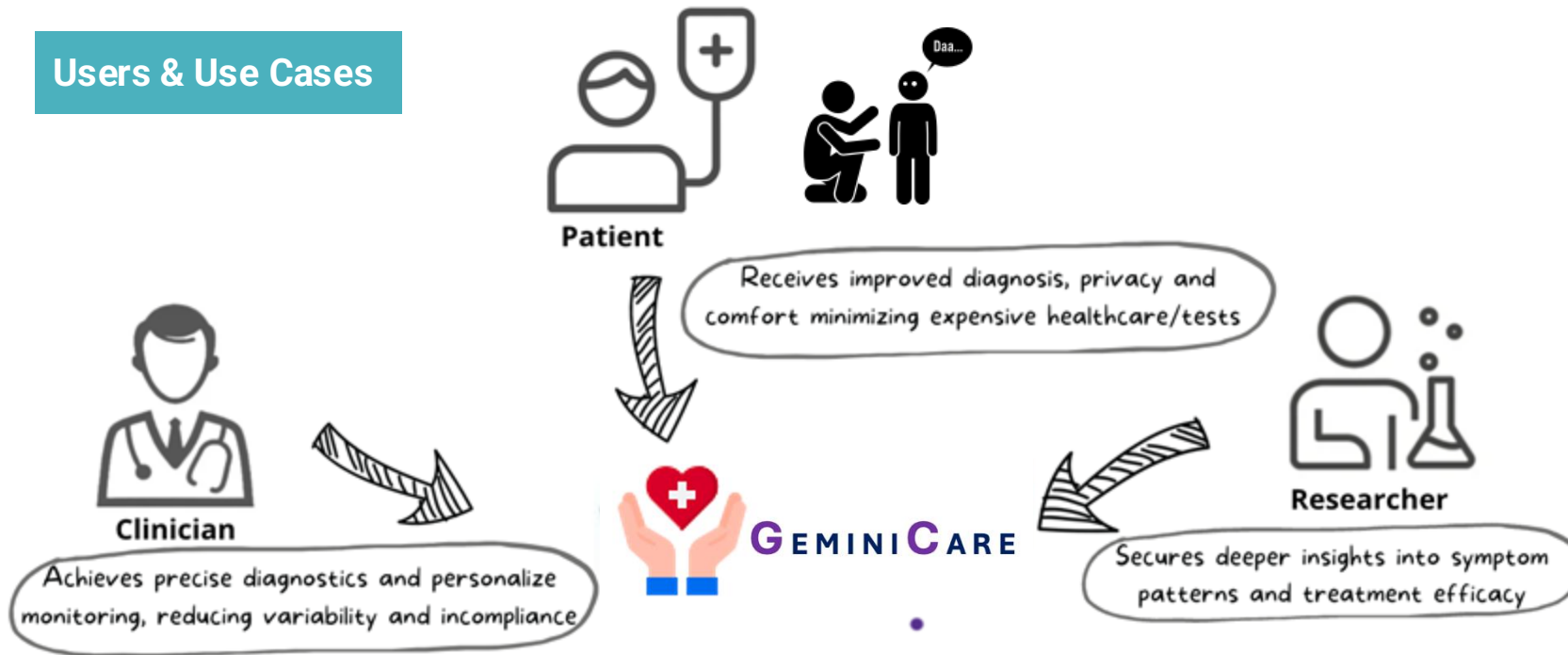
Reference: <https://www.autismparentingmagazine.com/autism-statistics/>

Solution

The primary goal is to develop a Multimodal Autism Monitoring & Analysis System that leverages Gemini AI to assist medical professionals by providing efficient method of diagnosing and monitoring autistic patients.

This application will analyze multimodal data inputs - video recordings, speech patterns, and textual information from patient interactions to detect and predict symptom patterns and severity, aiding medical professionals in making informed treatment decisions.

Users & Use Cases



Multimodal Data Analysis Use case – App Walkthrough

Clinician /
Researcher

Login Page



Gemini powered App for assisting Medical Professionals to diagnose, monitor and treat Autism patients efficiently.

Log in

Username

Password

Log In

Logs In

Home Page

Add Patient

Select an existing patient:

Ander

Patient Information for Ander

Fullname: *****

Age: 42

Gender: Male

Location: San Jose, CA

Medical History: History of Disorganized speech

Doctor Notes: (Date: Mar 15, 2024) Patient Ander, a 42-year-old male, has a history of depression. Current treatment regimen includes antidepressant medication and regular psychotherapy sessions. Patient's condition is stable with noticeable improvement in mood and overall well-being. Continued medication and therapy are recommended to maintain stability and prevent relapse.

Medical Tests Performed: MRI

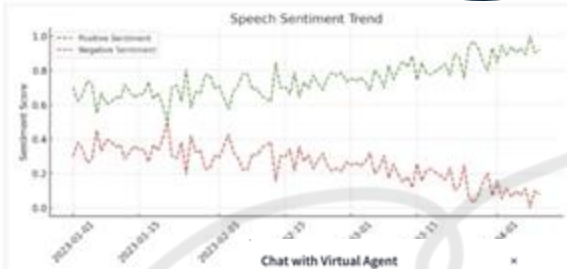
Family Members with Condition: 2

Update Profile

Perform Additional Diagnosis

Live Chat

NewSights Analysis:



View Patient
Statistics on
autistic
behavior

Live Chat
(based on
Knowledge
Base)
powered by
Gemini

Patient Diagnosis Page

Patient Data Analysis for Ander

Choose Input Type:

☒ Text
☐ Audio
☐ Video

By selecting Text, Audio or Video data about patient, I authorize the release of the information including my personal health record and diagnosis for medical treatment and research purposes.

Upload Text File

Drag and drop file here

Browse Files

speech_text_v1.docx

Text File Uploaded Successfully!

Analyze

View Content

Uploaded Text

"Everything's just too much sometimes, you know? I get so mad because the voices won't stop yapping. It's like they're yelling right into my ears and I can't shut them off. Makes me so tired. I told the doctor, but I don't think he gets it. Not really. It feels like nobody understands.

And the people staring doesn't help. They look at me funny, whispering behind my back. It just makes me angrier. I try to smile, show them I'm okay, but inside I'm screaming. Why can't they just leave me alone?

The shadows are the worst. They sneak around, almost catching me, and then they're gone when I look again. It scares me. I hold onto my usual things, my routines, because without them, I feel like I'd just float away into some dark place.

Visual Insights

Analyze Speech Text



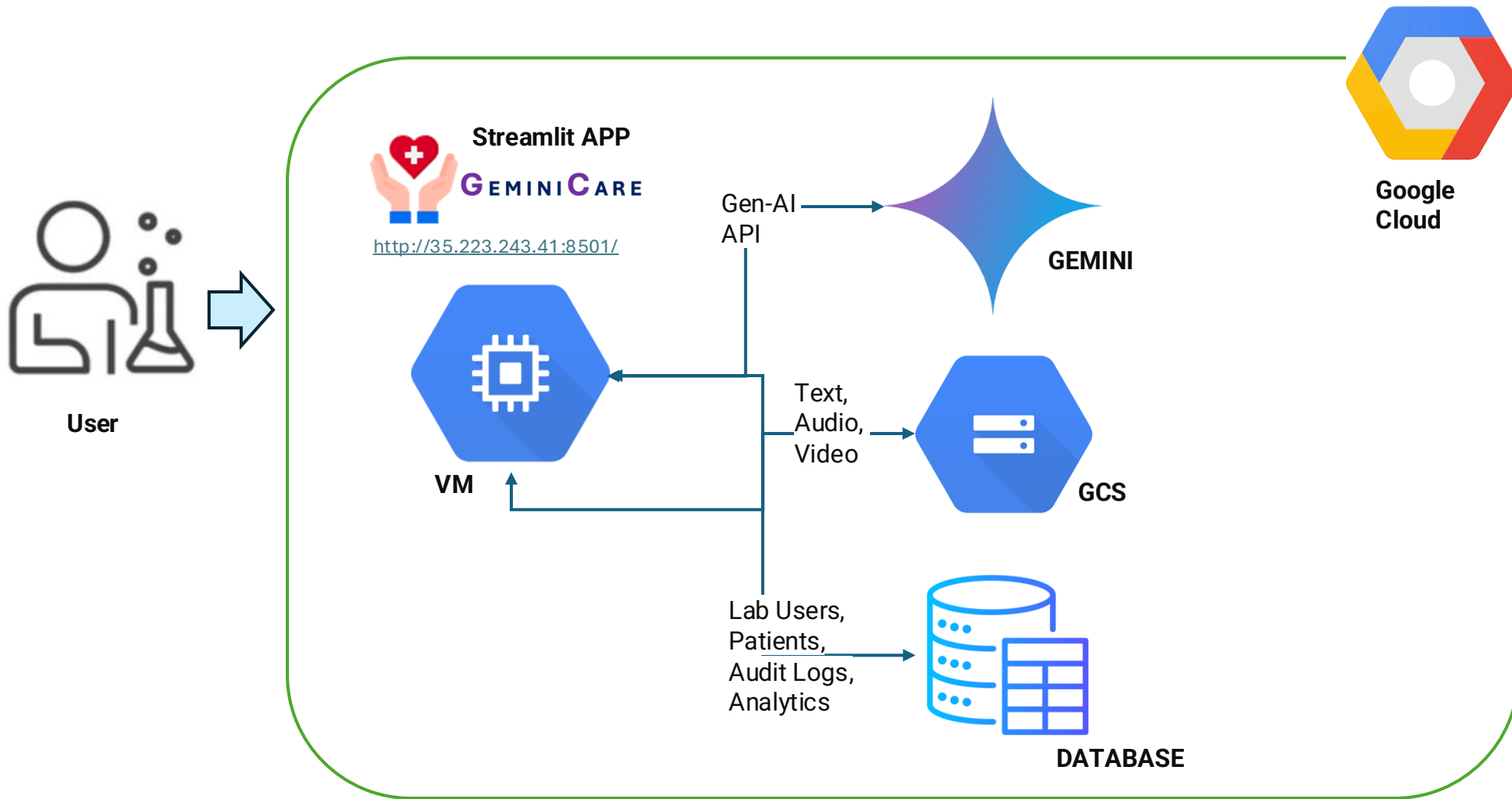
Patient Summary

This person's life has been a journey filled with struggles and triumphs as they navigate the complexities of living with schizophrenia. From a young age, he experienced feelings of isolation and misunderstanding, grappling with intrusive thoughts and overwhelming sensations that made daily tasks challenging. According to the new Newsights data and the data submitted on April 16th, 2024, as he entered adulthood, the symptoms of schizophrenia became more pronounced - the emotions primarily 'Anger' and 'Fear' index increased by 33% entering into adulthood, which may lead to hallucinations, delusions, and disorganized thinking dominating his reality.

Recommended Treatment: The patient needs therapy to re-establish courage to seek help, supported by their loved ones. The local community mental health teams (CMHTs) may also help. The medication treatment may bring some relief, but also may come with its own set of challenges, including medication side effects and the constant fear of relapse.

Upload patient's
info (text, audio
or video) ,
analyze
sentiments,
detect and
predict symptom
and severity

Deployment Diagram



Application of Generative AI



Patient Data Analysis for Ander

Choose Input Type:

- ☐ Text
☐ Audio
☒ Video

By selecting Text, Audio or Video data about patient, I authorize the release of the information including my personal health record and diagnosis for medical treatment and research purpose.

Upload Video File

Drag and drop file here
Limit 200MB per file • TXT, CSV, WAV, MP4, MOV, MPEG4

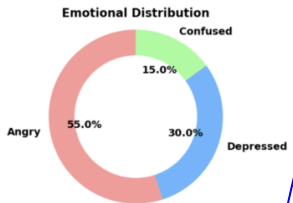
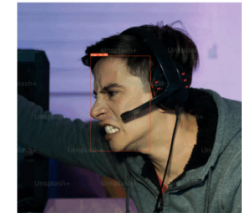
Browse files

Please Upload a Video File.

Analyze User Expression

☒ Visual Insights

Analyze Video Expressions



☒ Patient Summary

This person is suffering from Autism. His Geminicare video analysis on April 30th, 2024 suggests significant increase in emotions - 'Anger' and 'Depression' - bizarre characterized by emotional delusions, through disorders, visual and auditory hallucinations and paranoid ideations.

Recommended Treatment: The most effective interventions available are behavioral therapies based on applied behavioral analysis (ABA).

Live Chat

Multimodal Content Synthesis:

- **Interview Text, Speech Audio and Video Analysis:** Extensive sentiment analysis, tone analysis, intent keywords, trends with mental state.
- **Chain of Prompts:** Chain the Gen-AI responses to form new prompts and generate relevant insights with explanations.

Data Visualization and Analysis:

Data is presented using intuitive visualizations such as trend lines, scatter plots, and histograms, enabling clear tracking of progression or improvement over time. It is planned to automate insights and create charts using Gen-AI.

Application of AI for Prescriptive Analytics:

ML Algorithms will be used to cluster the insights and predict certain behaviour and generate recommendations.

AI-driven Knowledge Management:

RAG based Virtual Agent: It is planned to develop a RAG-based Agent which will assist the researchers and medical professionals by answering queries related to the patient data and broader schizophrenia research, facilitating an interactive, learning-enhanced environment.

Key Insights using Gemini Model

Speech Text

Everything's just too much sometimes, you know? I get so mad because the voices won't stop yapping. It's like they're yelling right into my ears and I can't shut them off. Makes me so tired. I told the doctor, but I don't think he gets it. Not really. It feels like nobody understands. And the people staring doesn't help. They look at me funny, whispering behind my back. It just makes me angrier. I try to smile, show them I'm okay, but inside I'm screaming. Why can't they just leave me alone? The shadows are the worst. They sneak around, almost catching me, and then they're gone when I look again. It scares me. I hold onto my usual things, my routines, because without them, I feel like I'd just float away into some dark place.



Sentiment: Negative		
Tone Analysis		
Tone	Score	Explanation
Anger	80	The text expresses strong anger towards the "voices" and the people staring. Phrases like "so mad," "yelling," "can't shut them off," and "makes me angrier" highlight this emotion.
Sadness	70	The speaker feels misunderstood and alone, leading to sadness. Phrases like "makes me so tired," "don't think he gets it," "feels like nobody understands," and "why can't they just leave me alone" convey this sadness.
Fear	60	The "shadows" and the feeling of "floating away into some dark place" evoke fear. The speaker is scared of the unknown and losing control.
Frustration	75	The speaker is frustrated by the inability to control the "voices" and the lack of understanding from others. Phrases like "won't stop yapping," "can't shut them off," and "I don't think he gets it" reflect this frustration.
Vulnerability	65	The speaker reveals their inner struggles and anxieties, making them appear vulnerable. Phrases like "I hold onto my usual things," "I feel like I'd just float away," and "It scares me" highlight this vulnerability.
Disappointment	55	The speaker is disappointed by the lack of support and understanding from others. Phrases like "I don't think he gets it" and "feels like nobody understands" convey this disappointment.
Loneliness	50	The speaker feels isolated and alone, longing for connection and understanding. Phrases like "feels like nobody understands" and "why can't they just leave me alone" suggest a sense of loneliness.

Keyword Analysis	
Intent Keyword	Explanation
Voices	This highlights the auditory hallucinations the speaker experiences.
Tired	This reflects the speaker's exhaustion from dealing with the voices.
Misunders tood	The speaker feels like their experiences are not understood by others.
Fear	The speaker expresses fear of the shadows and the judgment of others.
Control	The speaker clings to routines for a sense of stability and control.

Impact:

- **Improved Health Outcomes:** With more accurate diagnostics and personalized treatment plans, reduced symptoms, and improved quality of life.
 - **Social Inclusiveness:** standard unbiased diagnosis and treatment eliminate social stigma.
 - **Convenience:** privacy and comfort in treatment through virtual App
-
- **Research and development of new treatment methodologies:** The rich dataset generated by tool can be a valuable resource for ongoing research into autism. The rich dataset ensures accuracy, scale and diversity.
 - **Faster feedback loop:** improved connection with healthcare providers and patients enables faster research feedback loop.



- **Empowering Healthcare Providers:** GeminiCare uses AI to analyze complex, multimodal data, which minimizes the subjective bias inherent in traditional psychiatric evaluations. The standardized data driven diagnosis ensures timely identification of autism.
 - **Enhanced Monitoring and proactive treatment:** The continuous and non-invasive monitoring capabilities of the app allows for the detection of subtle changes in symptoms that might not be evident during routine clinical visits, enabling proactive management of the condition, potentially preventing severe episodes and reducing emergency interventions.
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- **Global Reach and Accessibility:** Given its digital and AI-driven nature, the app can be deployed across different regions, including underserved areas with limited access to specialized psychiatric services.
 - **Optimized National Health Expenditure:** Minimizes hospitalization and treatment cost managing the chronic mental health disorder. GeminiCare will enable better expenditure forecasting. It can also optimize healthcare resource utilization.
 - **Social Inclusiveness and productivity:** Recovered sociable and employable individual increases social value.

Ethical Guidelines



Fairness

The App is designed to provide equitable treatment recommendations regardless of a patient's background. The AI algorithms are internally trained by Google Gemini on diverse datasets to minimize biases related to age, gender, ethnicity, or socioeconomic status.



Reliability & Safety

AI technologies ensures accuracy and dependable outputs in clinical settings.

Patient names are anonymized/masked, and the recommendations are always reviewed and supplemented by human clinicians, ensuring patients' safety. The app flags data that deviates from norms.



Privacy & Security

GeminiCare team ensures that all services run securely in Google VPC and data is encrypted. User login will be strictly controlled and audited. User information will be used in AI Models, LLM and Analytics only based on user consent, giving users controls to choose how their medical data is used.



Inclusiveness

The tool caters to patients from diverse linguistic and cultural backgrounds by supporting multiple languages and adapting interfaces to different cultural contexts. This approach helps reduce healthcare disparities by making advanced healthcare technologies accessible to a broader population.



Transparency

We provide clear documentation and explanations of how the AI models operate, what data they use, and how decisions are made. The documentation ensures global explanation how the treatment recommendation works and local explanation of specific medication/therapy recommendation, further ensuring interpretability. This transparency ensures that clinicians understand the basis of the AI's recommendations, enabling informed decision-making.



Accountability

GeminiCare ensures that decisions made by the app can be audited and traced back to specific data inputs and AI processes. All operations performed by the user are logged. Regular performance reviews and updates are conducted to assess the impact of the AI, and stakeholders are involved in continuous dialogue to discuss outcomes and potential improvements.

Future Work



1

Need to provide secure user authentication and authorization.



2

Need to deploy the App in a scalable service (e.g. Cloud Run) with fail-over



3

Need to fix 'Lab User Sign Up' and 'Patient Registration' process



4

Need to store the uploaded files into GCS bucket seamlessly



5

Need to capture all the User Actions into a database table for auditing



6

Need to save all Analysis Results in order to show a trend of key metrics



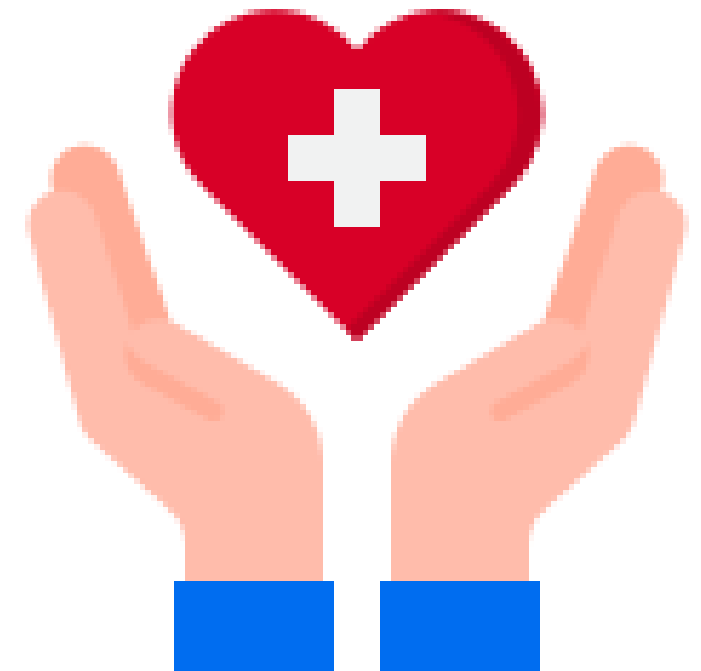
7

Need to keep the reports and historical data ready for the Clinician and Researchers



8

Need to implement the RAG-based Chatbot which can help implement Q&A on the knowledgebase and patient diagnosis details.



References

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